

**ALGEBRA I****Wednesday, June 19, 2019 — 1:15 to 4:15 p.m., only****Student Name** _____**School Name** _____

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

Print your name and the name of your school on the lines above.

A separate answer sheet for **Part I** has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 37 questions. You must answer all questions in this examination. Record your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in **Parts II, III, and IV** directly in this booklet. All work should be written in pen, except for graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale.

The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will *not* be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

Notice ...

A graphing calculator and a straightedge (ruler) must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

High School Math Reference Sheet

1 inch = 2.54 centimeters	1 kilometer = 0.62 mile	1 cup = 8 fluid ounces
1 meter = 39.37 inches	1 pound = 16 ounces	1 pint = 2 cups
1 mile = 5280 feet	1 pound = 0.454 kilogram	1 quart = 2 pints
1 mile = 1760 yards	1 kilogram = 2.2 pounds	1 gallon = 4 quarts
1 mile = 1.609 kilometers	1 ton = 2000 pounds	1 gallon = 3.785 liters
		1 liter = 0.264 gallon
		1 liter = 1000 cubic centimeters

Triangle	$A = \frac{1}{2}bh$
Parallelogram	$A = bh$
Circle	$A = \pi r^2$
Circle	$C = \pi d$ or $C = 2\pi r$
General Prisms	$V = Bh$
Cylinder	$V = \pi r^2 h$
Sphere	$V = \frac{4}{3}\pi r^3$
Cone	$V = \frac{1}{3}\pi r^2 h$
Pyramid	$V = \frac{1}{3}Bh$

Pythagorean Theorem	$a^2 + b^2 = c^2$
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Arithmetic Sequence	$a_n = a_1 + (n - 1)d$
Geometric Sequence	$a_n = a_1 r^{n-1}$
Geometric Series	$S_n = \frac{a_1 - a_1 r^n}{1 - r}$ where $r \neq 1$
Radians	1 radian = $\frac{180}{\pi}$ degrees
Degrees	1 degree = $\frac{\pi}{180}$ radians
Exponential Growth/Decay	$A = A_0 e^{k(t-t_0)} + B_0$

Part I

Answer all 24 questions in this part. Each correct answer will receive 2 credits. No partial credit will be allowed. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. Record your answers on your separate answer sheet. [48]

**Use this space for
computations.**

1 The expression $w^4 - 36$ is equivalent to

- (1) $(w^2 - 18)(w^2 - 18)$ (3) $(w^2 - 6)(w^2 - 6)$
 (2) $(w^2 + 18)(w^2 - 18)$ (4) $(w^2 + 6)(w^2 - 6)$

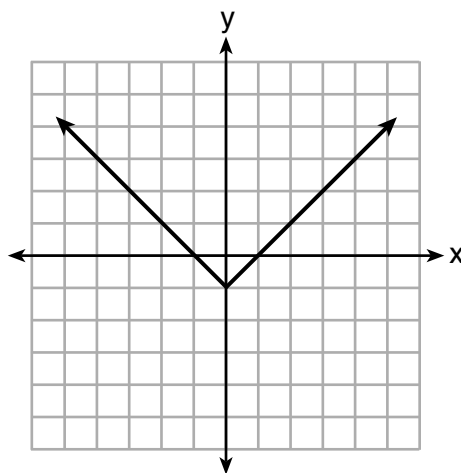
2 If $f(x) = 4x + 5$, what is the value of $f(-3)$?

- (1) -2 (3) 17
 (2) -7 (4) 4

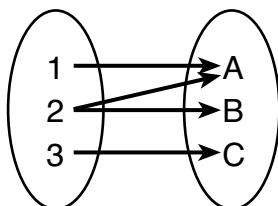
3 Which relation is *not* a function?

x	y
-10	-2
-6	2
-2	6
1	9
5	13

(1)



(3)



(4)

$$3x + 2y = 4$$

(2)

Use this space for
computations.

4 Given: $f(x) = (x - 2)^2 + 4$
 $g(x) = (x - 5)^2 + 4$

When compared to the graph of $f(x)$, the graph of $g(x)$ is

- (1) shifted 3 units to the left (3) shifted 5 units to the left
(2) shifted 3 units to the right (4) shifted 5 units to the right

5 Students were asked to write $6x^5 + 8x - 3x^3 + 7x^7$ in standard form.
Shown below are four student responses.

Anne: $7x^7 + 6x^5 - 3x^3 + 8x$
Bob: $-3x^3 + 6x^5 + 7x^7 + 8x$
Carrie: $8x + 7x^7 + 6x^5 - 3x^3$
Dylan: $8x - 3x^3 + 6x^5 + 7x^7$

Which student is correct?

- (1) Anne (3) Carrie
(2) Bob (4) Dylan

6 The function f is shown in the table below.

x	$f(x)$
0	1
1	3
2	9
3	27

Which type of function best models the given data?

- (1) exponential growth function
(2) exponential decay function
(3) linear function with positive rate of change
(4) linear function with negative rate of change

Use this space for
computations.

7 Which expression results in a rational number?

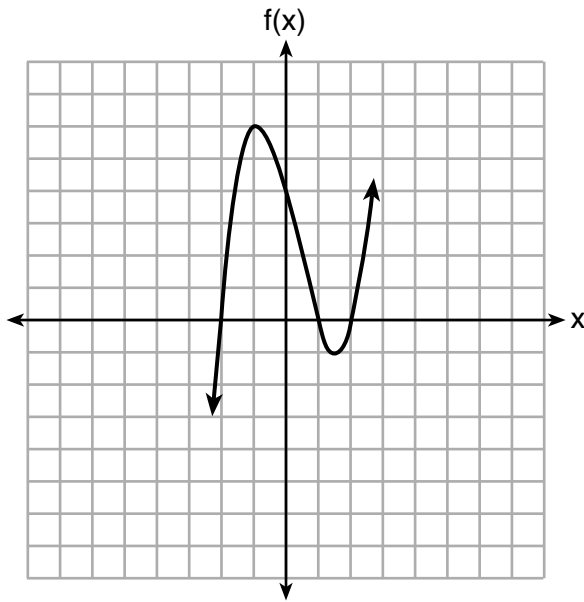
(1) $\sqrt{2} \cdot \sqrt{18}$

(3) $\sqrt{2} + \sqrt{2}$

(2) $5 \cdot \sqrt{5}$

(4) $3\sqrt{2} + 2\sqrt{3}$

8 A polynomial function is graphed below.



Which function could represent this graph?

(1) $f(x) = (x + 1)(x^2 + 2)$

(3) $f(x) = (x - 1)(x^2 - 4)$

(2) $f(x) = (x - 1)(x^2 - 2)$

(4) $f(x) = (x + 1)(x^2 + 4)$

9 When solving $p^2 + 5 = 8p - 7$, Kate wrote $p^2 + 12 = 8p$. The property she used is

(1) the associative property

(2) the commutative property

(3) the distributive property

(4) the addition property of equality

**Use this space for
computations.**

- 10** David wanted to go on an amusement park ride. A sign posted at the entrance read “You must be greater than 42 inches tall and no more than 57 inches tall for this ride.” Which inequality would model the height, x , required for this amusement park ride?

(1) $42 < x \leq 57$ (3) $42 < x$ or $x \leq 57$
(2) $42 > x \geq 57$ (4) $42 > x$ or $x \geq 57$

- 11** Which situation can be modeled by a linear function?

(1) The population of bacteria triples every day.
(2) The value of a cell phone depreciates at a rate of 3.5% each year.
(3) An amusement park allows 50 people to enter every 30 minutes.
(4) A baseball tournament eliminates half of the teams after each round.

- 12** Jenna took a survey of her senior class to see whether they preferred pizza or burgers. The results are summarized in the table below.

	Pizza	Burgers
Male	23	42
Female	31	26

Of the people who preferred burgers, approximately what percentage were female?

(1) 21.3 (3) 45.6
(2) 38.2 (4) 61.9

- 13** When $3a + 7b > 2a - 8b$ is solved for a , the result is

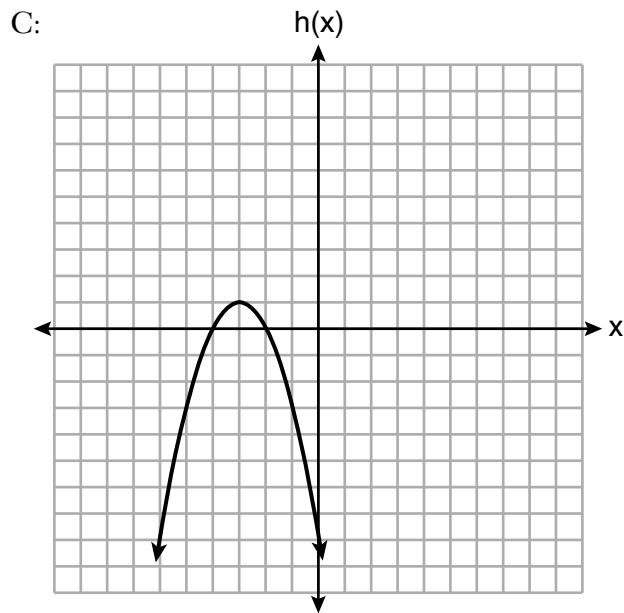
(1) $a > -b$ (3) $a < -15b$
(2) $a < -b$ (4) $a > -15b$

Use this space for
computations.

14 Three functions are shown below.

A: $g(x) = -\frac{3}{2}x + 4$

B: $f(x) = (x + 2)(x + 6)$



Which statement is true?

- (1) B and C have the same zeros.
- (2) A and B have the same y -intercept.
- (3) B has a minimum and C has a maximum.
- (4) C has a maximum and A has a minimum.

15 Nicci's sister is 7 years less than twice Nicci's age, a . The sum of Nicci's age and her sister's age is 41. Which equation represents this relationship?

- (1) $a + (7 - 2a) = 41$
- (2) $a + (2a - 7) = 41$
- (3) $2a - 7 = 41$
- (4) $a = 2a - 7$

Use this space for
computations.

- 16 The population of a small town over four years is recorded in the chart below, where 2013 is represented by $x = 0$. [Population is rounded to the nearest person]

Year	2013	2014	2015	2016
Population	3810	3943	4081	4224

The population, $P(x)$, for these years can be modeled by the function $P(x) = ab^x$, where b is rounded to the nearest thousandth. Which statements about this function are true?

I. $a = 3810$

II. $a = 4224$

III. $b = 0.035$

IV. $b = 1.035$

(1) I and III

(3) II and III

(2) I and IV

(4) II and IV

- 17 When written in factored form, $4w^2 - 11w - 3$ is equivalent to

(1) $(2w + 1)(2w - 3)$

(3) $(4w + 1)(w - 3)$

(2) $(2w - 1)(2w + 3)$

(4) $(4w - 1)(w + 3)$

- 18 Which ordered pair does *not* represent a point on the graph of $y = 3x^2 - x + 7$?

(1) $(-1.5, 15.25)$

(3) $(1.25, 10.25)$

(2) $(0.5, 7.25)$

(4) $(2.5, 23.25)$

- 19 Given the following three sequences:

I. 2, 4, 6, 8, 10...

II. 2, 4, 8, 16, 32...

III. $a, a + 2, a + 4, a + 6, a + 8...$

Which ones are arithmetic sequences?

(1) I and II, only

(3) II and III, only

(2) I and III, only

(4) I, II, and III

- 20** A grocery store sells packages of beef. The function $C(w)$ represents the cost, in dollars, of a package of beef weighing w pounds. The most appropriate domain for this function would be

(1) integers (3) positive integers
(2) rational numbers (4) positive rational numbers

- 21** The roots of $x^2 - 5x - 4 = 0$ are

(1) 1 and 4 (3) -1 and -4
(2) $\frac{5 \pm \sqrt{41}}{2}$ (4) $\frac{-5 \pm \sqrt{41}}{2}$

- 22** The following table shows the heights, in inches, of the players on the opening-night roster of the 2015-2016 New York Knicks.

84	80	87	75	77	79	80	74	76	80	80	82	82
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The population standard deviation of these data is approximately

(1) 3.5 (3) 79.7
(2) 13 (4) 80

- 23** A population of bacteria can be modeled by the function $f(t) = 1000(0.98)^t$, where t represents the time since the population started decaying, and $f(t)$ represents the population of the remaining bacteria at time t . What is the rate of decay for this population?

(1) 98% (3) 0.98%
(2) 2% (4) 0.02%

- 24** Bamboo plants can grow 91 centimeters per day. What is the approximate growth of the plant, in inches per hour?

(1) 1.49 (3) 9.63
(2) 3.79 (4) 35.83

Part II

Answer all 8 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [16]

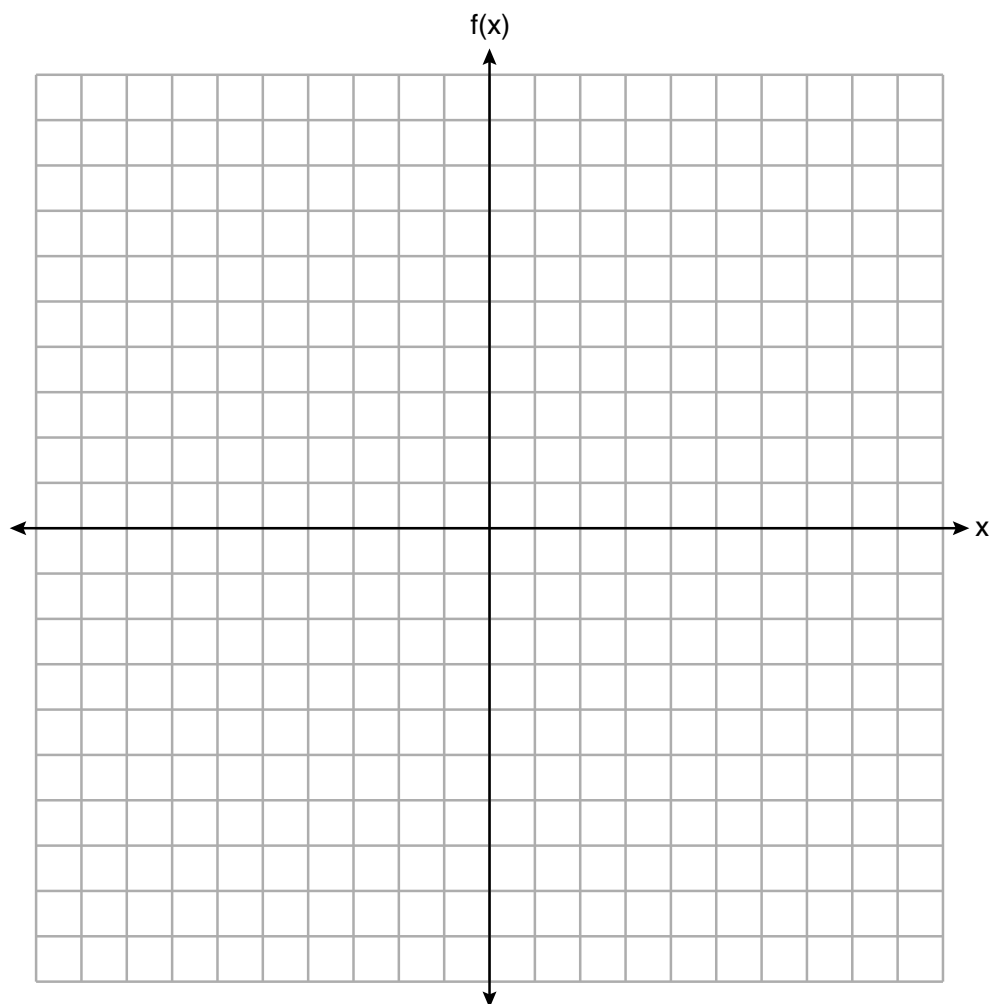
25 Solve algebraically for x :

$$-\frac{2}{3}(x+12) + \frac{2}{3}x = -\frac{5}{4}x + 2$$

26 If $C = G - 3F$, find the trinomial that represents C when $F = 2x^2 + 6x - 5$ and $G = 3x^2 + 4$.

27 Graph the following piecewise function on the set of axes below.

$$f(x) = \begin{cases} |x|, & -5 \leq x < 2 \\ -2x + 10, & 2 \leq x \leq 6 \end{cases}$$



28 Solve $5x^2 = 180$ algebraically.

- 29** A blizzard occurred on the East Coast during January, 2016. Snowfall totals from the storm were recorded for Washington, D.C. and are shown in the table below.

Washington, D.C.	
Time	Snow (inches)
1 a.m.	1
3 a.m.	5
6 a.m.	11
12 noon	33
3 p.m.	36

Which interval, 1 a.m. to 12 noon or 6 a.m. to 3 p.m., has the greatest rate of snowfall, in inches per hour? Justify your answer.

- 30** The formula for the volume of a cone is $V = \frac{1}{3}\pi r^2 h$. Solve the equation for h in terms of V , r , and π .

31 Given the recursive formula:

$$a_1 = 3$$

$$a_n = 2(a_{n-1} + 1)$$

State the values of a_2 , a_3 , and a_4 for the given recursive formula.

32 Determine and state the vertex of $f(x) = x^2 - 2x - 8$ using the method of completing the square.

Part III

Answer all 4 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [16]

- 33 A school plans to have a fundraiser before basketball games selling shirts with their school logo. The school contacted two companies to find out how much it would cost to have the shirts made. Company A charges a \$50 set-up fee and \$5 per shirt. Company B charges a \$25 set-up fee and \$6 per shirt.

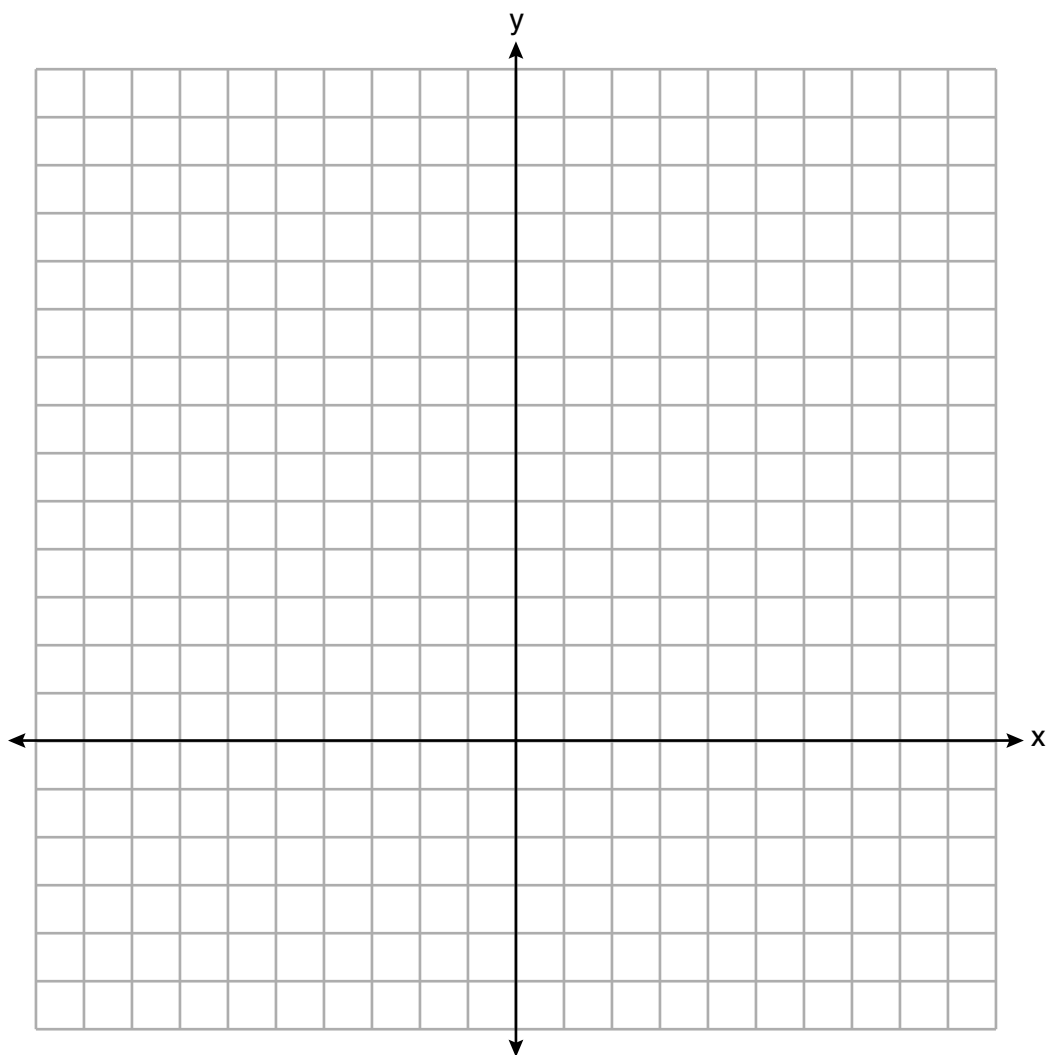
Write an equation for Company A that could be used to determine the total cost, A , when x shirts are ordered. Write a second equation for Company B that could be used to determine the total cost, B , when x shirts are ordered.

Determine algebraically and state the *minimum* number of shirts that must be ordered for it to be cheaper to use Company A.

34 Graph $y = f(x)$ and $y = g(x)$ on the set of axes below.

$$f(x) = 2x^2 - 8x + 3$$

$$g(x) = -2x + 3$$



Determine and state all values of x for which $f(x) = g(x)$.

- 35** The table below shows the number of hours ten students spent studying for a test and their scores.

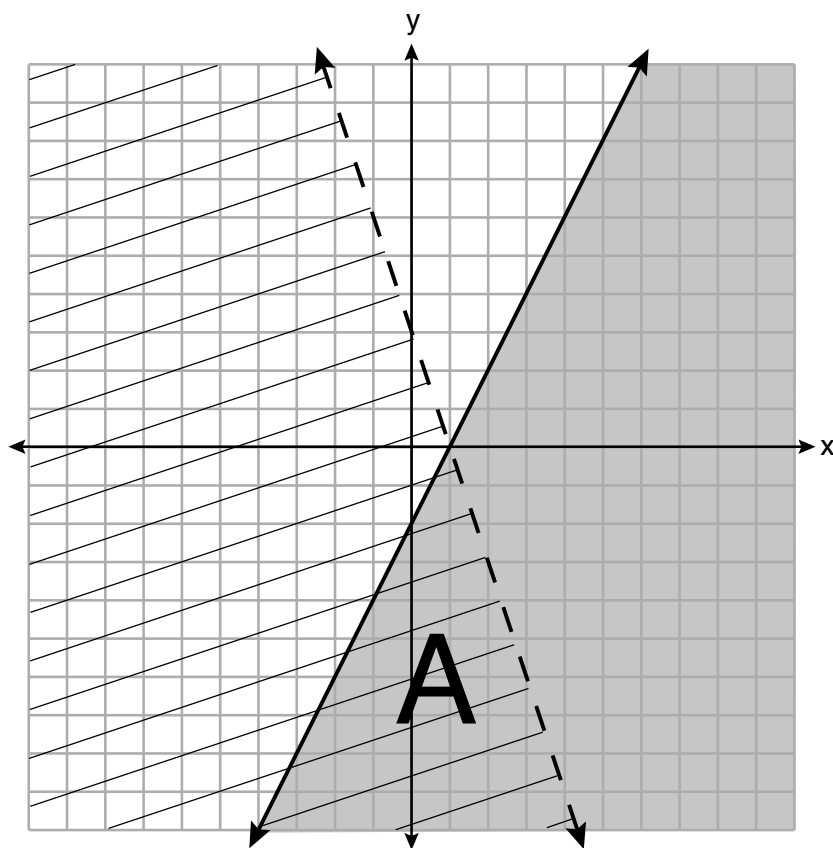
Hours Spent Studying (x)	0	1	2	4	4	4	6	6	7	8
Test Scores (y)	35	40	46	65	67	70	82	88	82	95

Write the linear regression equation for this data set. Round all values to the *nearest hundredth*.

State the correlation coefficient of this line, to the *nearest hundredth*.

Explain what the correlation coefficient suggests in the context of the problem.

36 A system of inequalities is graphed on the set of axes below.



State the system of inequalities represented by the graph.

State what region A represents.

State what the entire gray region represents.

Part IV

Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided to determine your answer. Note that diagrams are not necessarily drawn to scale. A correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [6]

- 37** When visiting friends in a state that has no sales tax, two families went to a fast-food restaurant for lunch. The Browns bought 4 cheeseburgers and 3 medium fries for \$16.53. The Greens bought 5 cheeseburgers and 4 medium fries for \$21.11.

Using c for the cost of a cheeseburger and f for the cost of medium fries, write a system of equations that models this situation.

The Greens said that since their bill was \$21.11, each cheeseburger must cost \$2.49 and each order of medium fries must cost \$2.87 each. Are they correct? Justify your answer.

Using your equations, algebraically determine both the cost of one cheeseburger and the cost of one order of medium fries.

Regents Examination in Algebra I – June 2019

Scoring Key: Part I (Multiple-Choice Questions)

Examination	Date	Question Number	Scoring Key	Question Type	Credit	Weight
Algebra I	June '19	1	4	MC	2	1
Algebra I	June '19	2	2	MC	2	1
Algebra I	June '19	3	4	MC	2	1
Algebra I	June '19	4	2	MC	2	1
Algebra I	June '19	5	1	MC	2	1
Algebra I	June '19	6	1	MC	2	1
Algebra I	June '19	7	1	MC	2	1
Algebra I	June '19	8	3	MC	2	1
Algebra I	June '19	9	4	MC	2	1
Algebra I	June '19	10	1	MC	2	1
Algebra I	June '19	11	3	MC	2	1
Algebra I	June '19	12	2	MC	2	1
Algebra I	June '19	13	4	MC	2	1
Algebra I	June '19	14	3	MC	2	1
Algebra I	June '19	15	2	MC	2	1
Algebra I	June '19	16	2	MC	2	1
Algebra I	June '19	17	3	MC	2	1
Algebra I	June '19	18	3	MC	2	1
Algebra I	June '19	19	2	MC	2	1
Algebra I	June '19	20	4	MC	2	1
Algebra I	June '19	21	2	MC	2	1
Algebra I	June '19	22	1	MC	2	1
Algebra I	June '19	23	2	MC	2	1
Algebra I	June '19	24	1	MC	2	1

Regents Examination in Algebra I – June 2019

Scoring Key: Parts II, III, and IV (Constructed-Response Questions)

Examination	Date	Question Number	Scoring Key	Question Type	Credit	Weight
Algebra I	June '19	25	-	CR	2	1
Algebra I	June '19	26	-	CR	2	1
Algebra I	June '19	27	-	CR	2	1
Algebra I	June '19	28	-	CR	2	1
Algebra I	June '19	29	-	CR	2	1
Algebra I	June '19	30	-	CR	2	1
Algebra I	June '19	31	-	CR	2	1
Algebra I	June '19	32	-	CR	2	1
Algebra I	June '19	33	-	CR	4	1
Algebra I	June '19	34	-	CR	4	1
Algebra I	June '19	35	-	CR	4	1
Algebra I	June '19	36	-	CR	4	1
Algebra I	June '19	37	-	CR	6	1

Key

MC = Multiple-choice question
CR = Constructed-response question

The chart for determining students' final examination scores for the **June 2019 Regents Examination in Algebra I** will be posted on the Department's web site at: <http://www.p12.nysed.gov/assessment/> on the day of the examination. Conversion charts provided for the previous administrations of the Regents Examination in Algebra I must NOT be used to determine students' final scores for this administration.